BTS – IV(SS) – 09.14 –0713 Reg. No. | | | | | | | | |

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B.Tech. Degree IV Semester Special Supplementary Examination September 2014

IT/CS 1405 DATA STRUCTURES AND ALGORITHMS

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer ***ALL*** questions)

(8 × 5 = 40)

I. (a) Differentiate between data structure, file structure and storage structure.
(b) What are the significance of hash functions?
(c) Explain the role of stack in postfix expression evaluation.
(d) Define a Red-Black tree.
(e) How are threaded binary trees represented in main memory?
(f) Discuss any one method used to represent graphs.
(g) Differentiate between B+ trees and B-trees.
(h) What do you mean by 'minimum spanning tree'? Explain any one method for construction of minimum spanning tree.

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PART B

(4 × 15 = 60)

II. Give the algorithm for quick sort. Mention its time complexity. Explain it with an example. (15)

OR

III. What are hash tables? Explain various hashing techniques. (15)

IV. (a) What are linked lists? Discuss about various categories of linked list. (7)
(b) Explain briefly how linked list can be used in polynomial manipulation. (8)

OR

V. (a) Write the translation rules for converting an infix expression to prefix expression. (10)
(b) What are sparse matrices? What are different representations? (5)

VI. (a) What are the characteristics of AVL trees. (8)
(b) What are threaded binary trees? Give its properties also. (7)

OR

VII. Explain briefly how linked lists can be used for representation of binary trees. Also discuss about the various binary tree traversal techniques. (15)

VIII. (a) Explain the Kruskal's algorithm. (7)
(b) Write an algorithm for finding the shortest path in path ADT. (8)

OR

IX. (a) Explain the various characteristics of a graph and how it can be represented using adjacency matrices and lists. (10)
(b) Explain DFS and BFS in detail. (5)

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